[Paper Critics 1] Design Tradeoffs for SSD Performance

1. Paper Overview

   This paper released from Microsoft played an essential role in the history of computer architecture. Before this paper, there was not enough research for internal architectures of SSD or the decision of SSD manufacturers to optimize the best efficiency. Depending on these issues, the paper raised the class of these design decisions and used these based on simulations of detected data collected from work balance in real system environments to analyze the likely performance in multiple settings. The paper figured out that SSD has high sensitivity from work balance for performance and lifetime and complex system problems that usually appear in the storage stack or even in distributed systems, which are related to equipment firmware.

2. Problem Definition

   Traditional design of operation system models based on HDD was met to exhibit significant limitations when faced with SSD. Data updates must be written to a new page and the old page marked as invalid, erased in block-level. This research cited data from the Samsung K9XXG08UM chip which takes only 25μs on page read but takes 200μs on page write and program. Erase block takes 1.5ms. This asymmetry resulting in severe long latency.

   Despite the extremely fast read speed for NAND, the 8-bit serial bus becomes a main performance issue and bottlenecks. Transferring a 4KB page to controller takes about 100μs, four times the reading speed of media. This data reveals why internal parallelism is the main reason for SSD.
   The random-access latency of traditional HDD is primarily dependent on mechanical seek time; however, SSD is primarily determined by clean performance. Misaligned writes, such as LBA mod 8 = 7, lead to additional page access, directly burdening the file system.

   These limitations force SSD controllers to implement an extremely complex layer which calls “FLT”. FLT must encapsulate garbage collection function and wear leveling logic in HW level. If SW engineers ignore this point, it would lead to write amplification, then cause to shutdown whole storage arrays of enterprise applications in a short time.

3. Methodology

   This paper uses DiskSim simulator to analyze the performance and test the marginal utility under different design parameters.

   The research deep compared the impact of mapping granularity for resource consumption. Page-level mapping provides optimal random writing performance, but it would consume very large amounts of RAM in the environment of HW at

that time. Conversely, Block-level mapping, although it could save space, it would lead to decrease two units of performance when read-modify-write frequently to deal with the random work balance which like TPC-C.

In order to conquer the bottleneck of sequential interconnects, author proposed several parallel strategies: interleaved interconnect and parallel interconnect which include shared bus parallel interconnect and shared control parallel interconnect.

For interleaved interconnect, it used multiple planes within the chip to sequential transfers and program control operations in pipeline. For parallel interconnect, including shared bus and control parallel interconnect. The shared bus one would increase capacity without increasing bandwidth, suitable in low-cost situations. Another one, shared control would implement synchronous operations through independent data paths. It needs to be decided between these two interconnect modes.

It also launched another design for SSD that the research appears that reverse 15% of physical spaces as the overprovisioning is the perfect balance between business and technology. This mechanism ensures that the garbage collector still could have enough space to work in a high-loading situation, that could alleviate congestion in front-end.

According to paper, it also raised a very smart approach for cold data migration to decrease the consumption of the SSD block. In traditional method to reclaim block by greedy algorithm which prioritized the most invalid pages. It would lead to some hot data blocks ageing rapidly because of frequent reclamation, while the blocks stored cold data erased infrequently. To solve this problem and balance the consumption for each block, the paper raised an algorithm is that if one of the blocks ages less than a threshold, then trigger the algorithm to move the cold data to that block. It means that it could decrease the rate of consumption of specific block due to cold data doing I/O infrequently.

4. Prons and Cons

For Prons in my opinion, the author released an approach to standardize SSD architecture with a complete design framework for SSD developers. The most important attribution is that constructs 4 main dimensions: Data Placement, Parallelism, Write Ordering and Workload Management. In my opinion, the most attractive and interesting concept it raised is that use the cold data migration strategy instead of greedy algorithms to avoid the consumption of blocks. In this design, it only wasted 4.7% latency then achieved the result with 0 expired block. That’s the most amazing work I think for pros.

One the other hand, for Cons in SSD design, due to physical restriction of NAND flash memory in rewrite, delete and age issues, the low-level FW and HW development engineer need to afford very huge design tasks on these. It would take a lot of time to maintain system for FW engineers. Another con I want to emphasize is that since maintaining memory age and efficiency of SSD, it had a design principle “overprovisioning” to decrease consumption and balance the usage for blocks in memory. It would cost a little out-of-control latency to migrate cold data. So, in this case, the threshold of block needs to be set up very accurately to have the most excellent balance between these two restrictions.

5. Future Work in My Opinion

Future work in my opinion. The first one I want to say is I figure out that the design principles are very clear, but for designing problems for wear-level optimization I mentioned in second con that I think we could design more specific algorithms to define the threshold. In the paper, it set it by a static number in SSD to trigger the migration, If I could design a more convenient algorithm which can dynamically define the threshold. That could make the threshold dynamic decided by the different situations the SSD hardware actually meet to optimize the usage and balance of blocks’ consumptions. Furthermore, it also could deal with some problems which appear in a very emergency that need to backup or migration immediately issues in the future.

Second, for the simulator DiskSim. I think that the functions of DiskSim for SSD are like the compiler for programming languages. In compiler field has a very huge development issues is that if translate the code to hardware design could reserve any object or file-flop to implement the logic components then it could decrease the cost of products brought a lot of profits for companies. So, in the similar case here, I think that for designing these SSD applications if the DiskSim could translate to more efficient or more friendly transformation to them. I think it could have more potential for SSD design area.